

Project Design Phase

Module 2: Proposed Solution

Project Name: Import Data Using Transform Maps (Spreadsheet)

1. Executive Summary

The proposed solution introduces an automated, scalable, and reliable data ingestion system leveraging an **Import Set and Transform Map Engine**. This solution eliminates the manual effort and error-prone nature of re-keying spreadsheet records into centralized target database tables. By staging incoming spreadsheet files (.xlsx or .csv) in an intermediate layer before applying strict transform mapping rules, the system ensures high data accuracy, automated record deduplication, and full operational traceability.

2. Core Features & System Capabilities

- **Automated Spreadsheet File Parsing:**
Supports uploading standard Microsoft Excel (.xlsx) and Comma-Separated Values (.csv) formats. The engine automatically parses headers and creates corresponding fields in a temporary staging area.
- **Intermediate Staging Layer (Import Set Table):**
Acts as an isolated sandbox holding uploaded raw data before committing to production tables. This prevents bad, incomplete, or corrupted records from affecting target system data.
- **Flexible Field Mapping & Transformation Rules:**
 - **Auto-Map Matching:** Automatically matches source spreadsheet column names to destination table fields when names match.
 - **Mapping Assist:** Provides a graphical interface to map custom spreadsheet headers to target table fields.
 - **Transform Scripts:** Enables custom JavaScript logic (e.g., onBefore, onAfter) to cleanse, format, or calculate values dynamically during the import process.
- **Smart Deduplication via Coalesce Fields:**
Allows administrators to set unique identifier fields (such as *Employee ID*, *Email*, or *Asset Number*) as Coalesce = True.
 - If a matching value exists in the target table, the system **updates** the existing record.
 - If no match exists, a **new record** is inserted.
- **Auditability & Execution Logging:**
Generates comprehensive import set logs detailing inserted rows, updated rows, ignored rows, and error records for instant troubleshooting.

3. Proposed End-to-End Workflow

1. **Source File Upload:**

The user or system administrator uploads the source spreadsheet containing bulk data.

2. **Import Set Generation:**

The platform creates an intermediate Import Set Table (staging table) and populates it with raw spreadsheet data.

3. **Transform Map Configuration & Execution:**

The system applies pre-configured field mappings, runs transform scripts, and checks Coalesce key rules.

4. **Target Table Commit:**

Processed and validated records are safely committed to the destination database table, and an execution summary log is generated.

4. Expected Benefits & Outcomes

- **80%+ Reduction in Processing Time:** Eliminates hours of manual data entry for large datasets.
- **100% Schema Accuracy:** Direct mapping prevents data placement errors across fields.
- **Zero Record Duplication:** Coalesce logic maintains single-source-of-truth database integrity.